

From Reactive to Proactive: How Intuit Credit Karma Solved Data Quality at Scale

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Intuit Credit Karma

Session Agenda

groups

Introduction

About Us & Team Mission

view_quilt

Quality Pillars

Data Frameworks & Trust

psychology

AI Automation

Scaling with Intelligence

map

Learnings

The Quality Journey

play_circle
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Live Demo

System in Action

forum

Q&A

Open Discussion

Designed Alliance, what is this session **NOT** about?

- Not a Beam/Dataflow pipeline-internals deep dive
 - No windowing, triggers, DoFns, or IO-connector tuning
- No ML Model Quality or Explainability
- Not a MonteCarlo product tutorial or vendor pitch
 - We do leverage MonteCarlo and it's helped us tremendously in our Data Quality journey
- No pipeline performance/throughput/cost optimization

Designed Alliance, what is this session about?

- A repeatable framework - 5 pillars – Timeliness, Completeness, Accuracy, Observability, Governance – that turn "is our data & Apache Beam pipeline trustworthy?" into something you can measure and enforce.
- The honest journey, scars included, ouch!
 - What worked,
 - What we'd do differently, and
 - How we fought alert fatigue and sensitivity decay to land on high-confidence, low-noise paging.
- Live Demo of AI skills in action (and by that we mean, a *Recorded Video* ;)

About Intuit Credit Karma

Intuit Credit Karma

Our Mission

Champion financial progress for all – giving 140M+ members the tools, insights, and confidence to reach their financial goals.

What Makes Us Different

Pioneered the free credit score model. Our core services remain free – no credit card required, no hidden fees.

What We Offer

Credit & identity monitoring | Credit cards & loans | Insurance | Savings & checking accounts



About Credit Ecosystem Team

The Credit Ecosystem team is the data backbone of Credit Karma – we are the gateway to ingest, transform, and serve the financial data that powers critical business functionalities and needs.

Data Ingestion

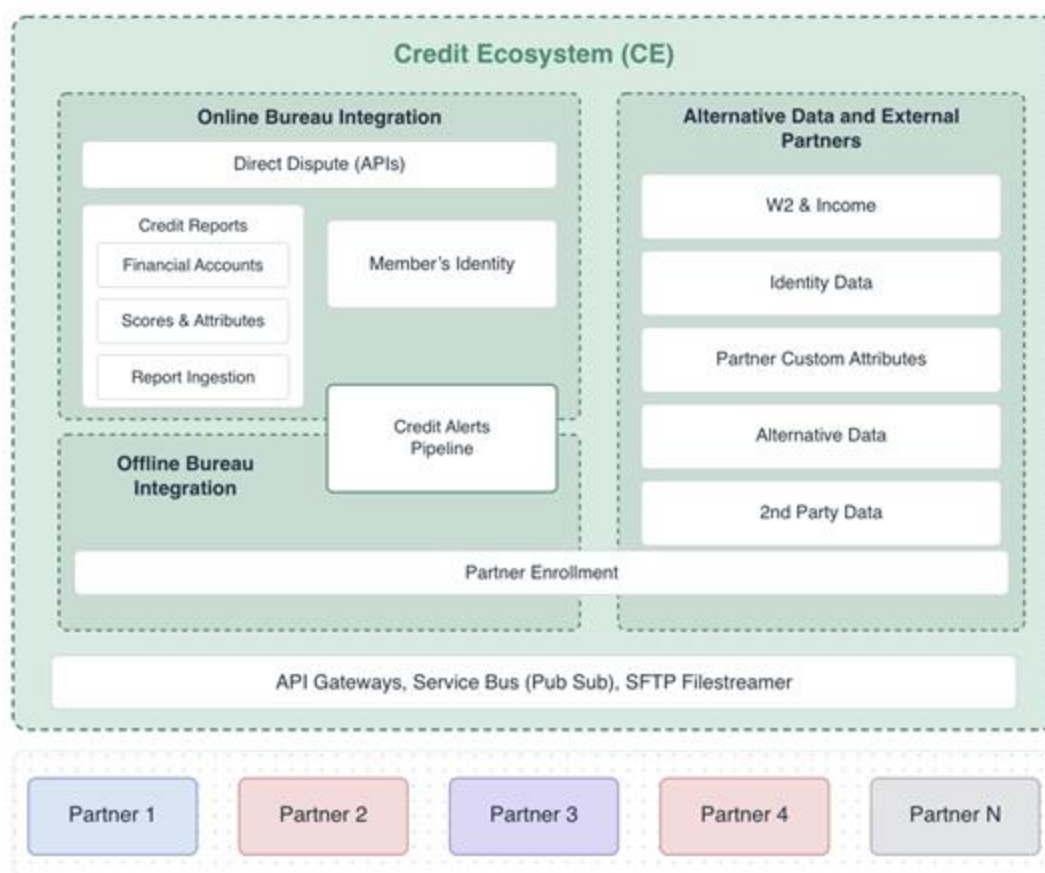
First point of contact for all bureau and partner data – handling real-time streams and batch transfers.

Broad Service Ownership

100+ data tables powering 20+ internal services across 7+ distinct data domains.

Single Source of Truth

Building the unified model of our members' financial profiles.



About Us

Puneet Singh

Senior Data Engineer, Credit Ecosystem

12+ years in Data Engineering and Data Quality. Career spanning Teradata, Hadoop/Spark, and cloud-native microservices on AWS and GCP – with deep expertise in bureau data pipelines.



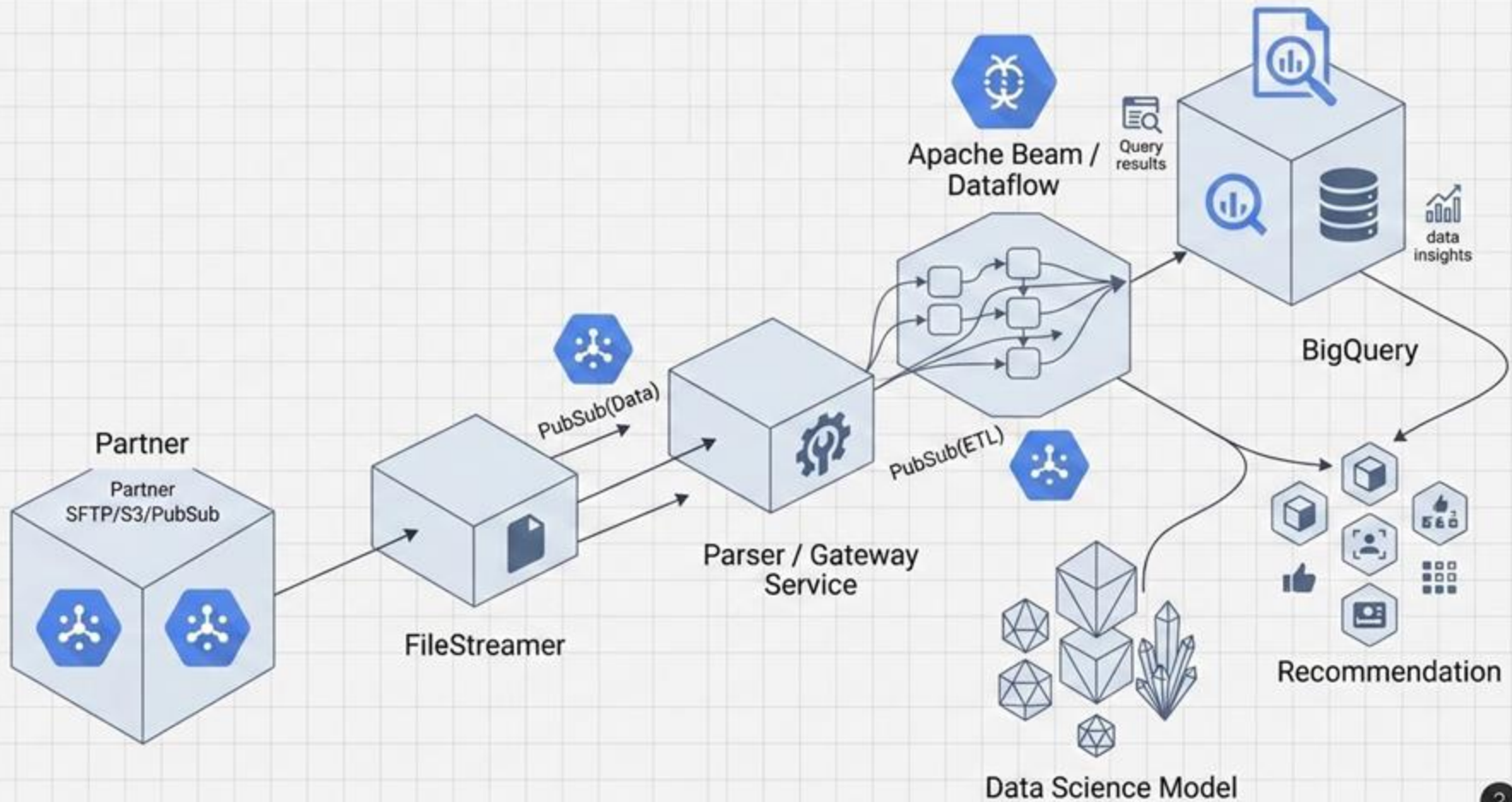
Veenit Shah

Senior Data Engineering Manager, Credit Ecosystem

Data engineering leader with 14+ years spanning individual contributor to Engineering Manager. I currently lead **Credit Ecosystem** – a core, mission-critical engineering team at Credit Karma. Focused on building high-scale data pipelines and growing high-performing teams.



Data Pipeline Overview



Problem Statement

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Data Scale & Complexity

100+ tables, 40K+ columns, and 140M members.

Challenge: What is the priority?

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Service Reliability

`java.net.SocketTimeoutException: connect timed out`

Impact of one profile exception across 30+ services and 20+ data pipelines.

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Murphy's Law in Action

"Anything that can go wrong, will go wrong... and it did!"

Lack of controls for bad data ingested within CK.



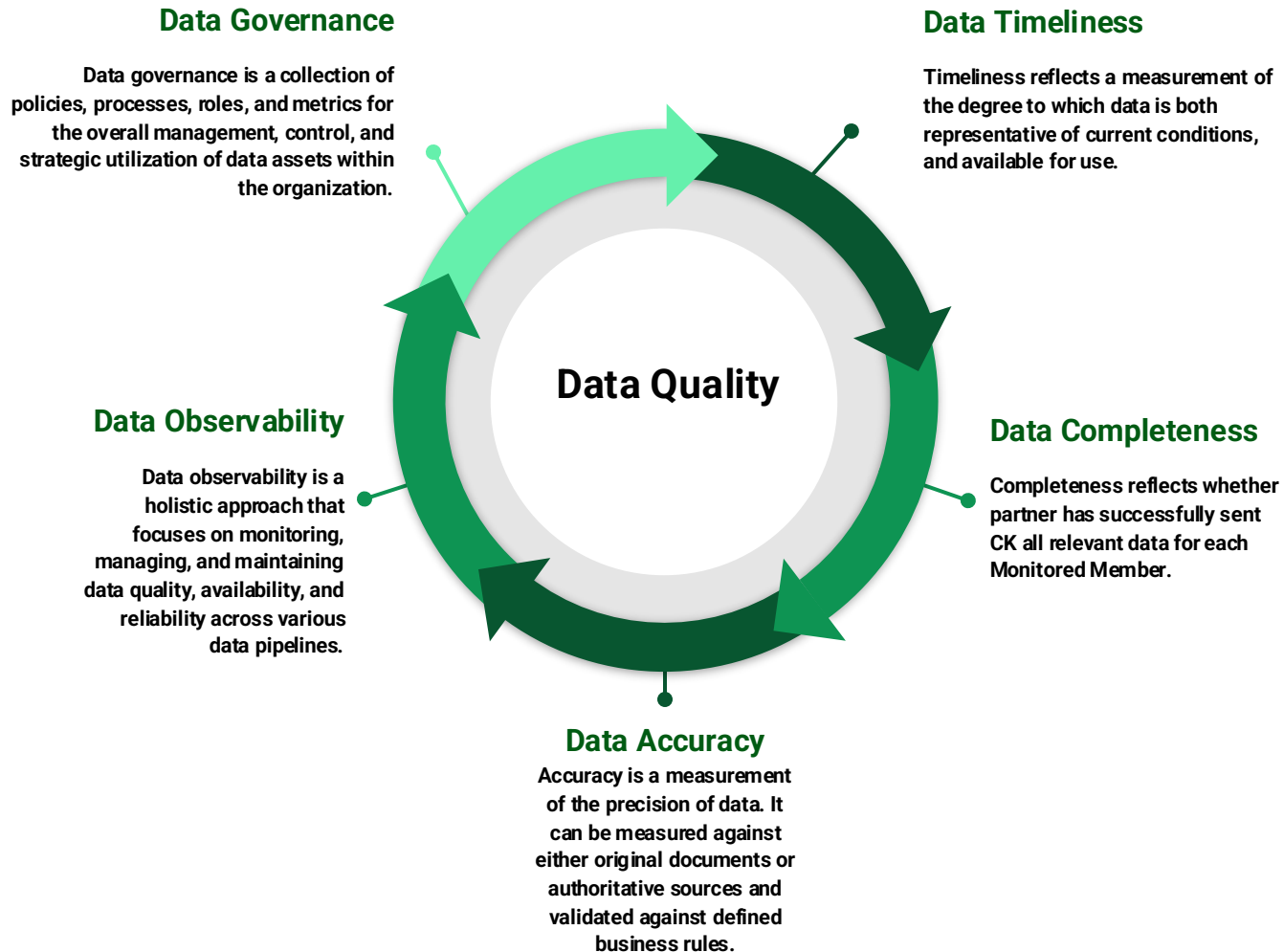
AS YOU CAN SEE, OUR TOP MARKETS ARE
UNITED STATES, CANADA, USA AND THE U.S.

Misalignment and lack of visibility

"Our top markets are United States, Canada, USA and the U.S."

Data Quality Pillars

Data Quality Pillars



Data Timeliness

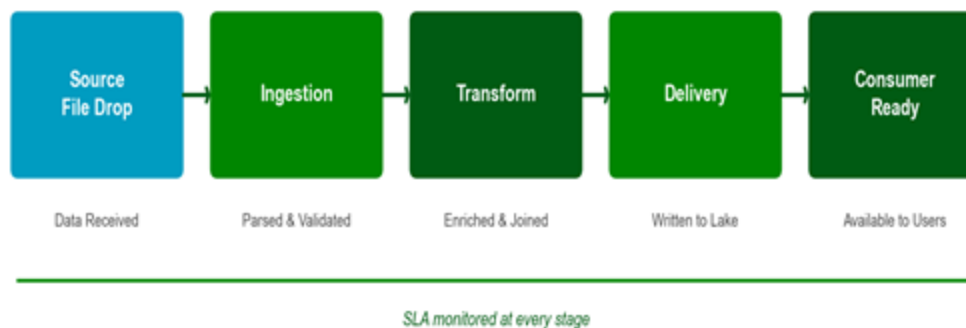
What We Measure

- SLA timestamps captured at every pipeline hop – ingestion, transformation, delivery
- End-to-end latency tracked from source receipt to downstream availability
- SLA contracts defined per partner and pipeline (e.g., daily bureau files by 08:00 UTC)
- Automated alerts fire when any stage breaches SLA threshold

Key Learnings

- Start with your most critical pipeline – prove value, then expand iteratively
- Hold data sources accountable with contractual SLAs, not just internal targets
- End-to-end visibility builds stakeholder trust faster than point-in-time checks

End-to-End SLA Tracking



Data Completeness

How We Enforce Completeness

- Metadata pipelines compute statistical profiles per batch: counts per file/batch, min, max, std dev, null rates, and special value frequencies
- Data contracts and data dictionaries define authoritative expected ranges for every key field
- Automated checks compare incoming data against these contracts – deviations(>99.5%) trigger alerts
- Credit score range validation, date sanity checks, and cross-bureau consistency checks run daily

Manifest Completeness Model

Expected Records

140,000,000 (100%)

99.5%

Alert Threshold: < 99.5%

Real Incidents Caught

- 1 in 40 partner files silently failed
- Duplicate file drops causing inflation

Key Learnings

- Manifest contracts are low effort, high reward
- Record-count validation is the fastest path

Data Accuracy

How We Enforce Accuracy

- Metadata pipelines compute statistical profiles per batch: min, max, std dev, null rates, and special value frequencies
- Data contracts and data dictionaries define authoritative expected ranges for every key field
- Automated checks compare incoming data against these contracts – deviations trigger alerts
- Credit score range validation, date sanity checks, and cross-bureau consistency checks run daily

Key Learnings

- No single rule fits all – accuracy checks are domain-specific and require expert input
- Prioritize high-ROI alerts (credit scores, income figures) over exhaustive coverage
- Balance thoroughness with alert fatigue – alert on deviation, not just absence
- Accuracy monitoring is the most resource-intensive pillar – invest incrementally

HIGH IMPACT / LOW FREQ

Monitor Selectively

Deep dives into core metrics like avg. value spikes in key columns.

HIGH IMPACT / HIGH FREQ

Page On-Call

Mission critical values outside data dictionary ranges. Immediate action required.

LOW IMPACT / LOW FREQ

Log Only

Minor field variance captured in metadata pipelines for audit.

LOW IMPACT / HIGH FREQ

Reduce Noise

Handle null rate fluctuations through automated baseline adjustments.

Data Observability

- **The Challenge:** 100+ tables, 40K+ columns — manual checks impossible
- **Solution:** Monte Carlo deployed as enterprise data observability platform
- **Result:** From reactive firefighting → proactive observability

Observability Architecture



Integrated with PagerDuty for real-time on-call alerting · Asset Registry: owner, priority (P0-P3), SLA, DQ rules in YAML

Data Governance

The Problem

Hundreds of data assets, no centralized ownership, inconsistent alerting.

The Solution: Data Asset Registry

Every data table registered as a governed asset – defined in code (YAML), versioned, and reviewed.

P0 — Mission Critical

Action: Page on-call immediately

P1 — High Priority

Action: Slack alert within 15 minutes

P2 — Standard

Action: Email notification

P3 — Low Priority

Action: Log & review weekly

Defined in Code

Versioned YAML files define the source of truth for every table.

SLA Thresholds

Codified freshness and volume targets per asset tier.

Schema Contracts

Field-level validation enforced by automated DQ monitoring.

Alert Automation leveraging AI

Problem: Automate 1000+ lines of DQ platform effort

Solution: AI Driven DQ automation

How it works: Technical/Non-technical users invoke partner-dq skill → auto-generates PR's across different repos → Review(human in the loop) → Deploy in production

partner-dq skill

Define the dataset once
(intake doc)

→ Let AI build DQ platform

1

Data Quality Context

Data Quality Standards
Data Quality Guidance
Platforms/Repos patterns

2

Completeness & Accuracy

Completeness
Dictionary-driven accuracy
Compliance rules

3

Data Observability & Timeliness

NewRelic Service metrics/alerting
Auto-onboard to anomaly detection
Field health: nulls + volume
Timeliness for non-daily feeds

4

Data Governance

Data Asset Registry
Schema
Data Owners / Business Owners
Star Rating

Alert Triaging with AI

The Problem

200+ monitoring checks across all data partners – each alert sends on-call to runbooks, requiring 15-45 min of manual diagnosis before any action.

Our Solution

An AI-powered diagnostic layer that sits on top of existing alerts – zero changes to current infrastructure.

Result: Root cause + recommended action delivered in ~12 seconds vs. 45 minutes.



Demo : New Pipeline Onboarding & Alert Triaging leveraging AI

Should I also generate clean-schema docs for the Beam table (the generate_clean_schema part)? (Yes, via CleanData / No, DQ only)

/ce-plugin:ce-bq-context Use skill /partner-dq to generate DQ for Beam partner dataset. Refer to data dictionary https://docs.google.com/spreadsheets/d/1MM4xF00Y0NJ_qqOE1Ud8mn2ms5momZAu6oTyG02EKgs/edit?gid=99681872#gid=99681872 and the table will be prod-ck-abl-data-53.ck_bureau_ingestion_etl_0_2.beam_attributes_usa. Once all the changes made give me stats of how many PR's needs to be opened, total code lines written, etc. Also leverage generate_clean_schema skill. Get out of existing skills and start fresh using /ce-bq-context

- You've re-invoked /ce-bq-context and asked me to start fresh. I now have the CE BigQuery context loaded. Let me restate the task and plan clearly.

Task understanding

Goal: Generate Data Quality (DQ) monitoring for the Beam partner dataset, specifically the table:
- prod-ck-abl-data-53.ck_bureau_ingestion_etl_0_2.beam_attributes_usa

Inputs:
- Data dictionary (Google Sheet): 1MM4xF00Y0NJ_qqOE1Ud8mn2ms5momZAu6oTyG02EKgs, gid 99681872
- Skills requested: /partner-dq and generate_clean_schema

Deliverable: DQ artifacts + stats (PRs to open, total lines of code written, etc.)

What I need to flag before doing the work

I want to be upfront so I don't burn effort in the wrong direction:

- The two named skills (/partner-dq, generate_clean_schema) are not installed in this environment. I searched the skill loader and tool index – neither resolves. The closest installed equivalents are:
 - DQ generation: the ce-plugin:ce-dqm-* family (ce-dqm-list, ce-dqm-details, ce-dqm-run, ce-dqm-debug) – these produce DQM alert configs (YAML + SQL completeness/accuracy checks) that get PR'd into the ce_corp-composer repo. This is the CE-team mechanism that matches "generate DQ for a partner dataset" and your "how many PRs / lines of code" question.
 - Clean schema: ck-bi:ck-bi-cleandata (CleanData) – authors table/column documentation (clean schema).

/ce-bq-context (now loaded) is context-only – it tells me about CE BigQuery tables but does not itself generate DQ. Notably, beam_attributes_usa is not in its table index, so Beam is a newer/partner table I'll need to discover the schema for directly.

- The data dictionary is a Google Sheet – I'll need the gdrive MCP (read_sheet / list_sheet_tabs) to read it. That tool is available.

I'm not going to silently substitute skills or invent a /partner-dq workflow from memory, because the output (real YAML/SQL files – real PRs) and the stats you asked for depend entirely on which mechanism is the "correct" one for your team. Let me confirm the one decision that changes everything, then I'll execute end-to-end without further interruption.

- User answered Claude's questions:
 - Do you want the clean-schema step (generate_clean_schema) as well, and via what? – Yes – via CleanData
- You answered the clean-schema question (Yes – via CleanData) but the first question (the DQ mechanism – which is the one that determines everything) wasn't answered. That's the decision I can't proceed without, so let me ask just that one.
- User answered Claude's questions:
 - For the DQ generation itself: the named /partner-dq skill isn't installed here. Which mechanism should I use to generate DQ for beam_attributes_usa? (Clean schema is settled – you chose CleanData.) – Use ce-dqm-* (composer)
- Decisions locked in:
 - DQ: ce-dqm-* (composer) – author YAML + SQL DQM alerts for beam_attributes_usa, PR into ce_corp-composer

Challenges & Learnings

CHALLENGES

ALERT FATIGUE

Overexposure to information leading to decision overload. Critical information is often missed or ignored.

REDUCED SENSITIVITY

Increased threshold for response. Genuine issues require higher alert levels to be noticed.

EMOTIONAL DISSOCIATION

Burnout and disengagement. Lack of care for consequences due to exhaustion.

UNINTENDED OPERATIONAL DISRUPTIONS

Inefficient processes and poor decisions. Fixes create new, overlooked problems.

LEARNINGS

THRESHOLD-BASED PRIORITIZATION

Utilize a risk-score threshold to only escalate high-priority items to reduce noise.

CONTEXTUAL FILTERING

Implement automated context checks to reduce false positives effectively.

OPTIMIZED ALERT QUALITY

Focus on clear, actionable instructions in alerts to improve response times.

CROSS-FUNCTIONAL WORKFLOWS

Integrate cross-team data streams to align understanding of underlying issues.

3 takeaways from this session . . .

1

Know your data

- Each partner
- Each table
- Each column
- Metadata dictionaries are your friend

2

Measure to improve

If you can't measure it, you can't improve it.

- Establish baselines first
- Identify current state

3

POC first, pivot

Monitor 10k+ columns effectively.

- Prioritize tasks
- Gather feedback
- Pivot as needed

Thank you

Puneet Singh

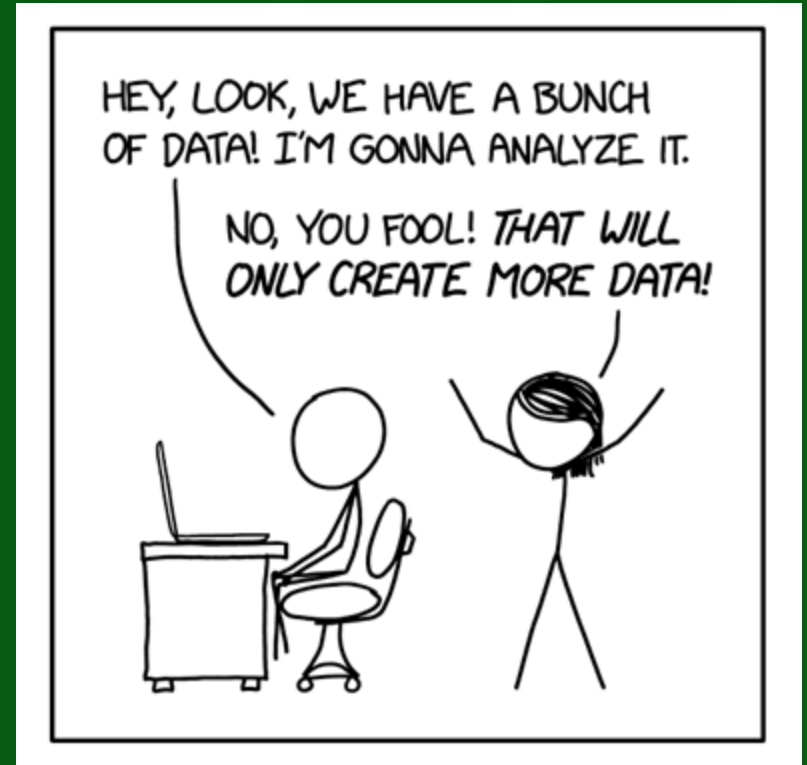
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<https://xkcd.com/2582/>